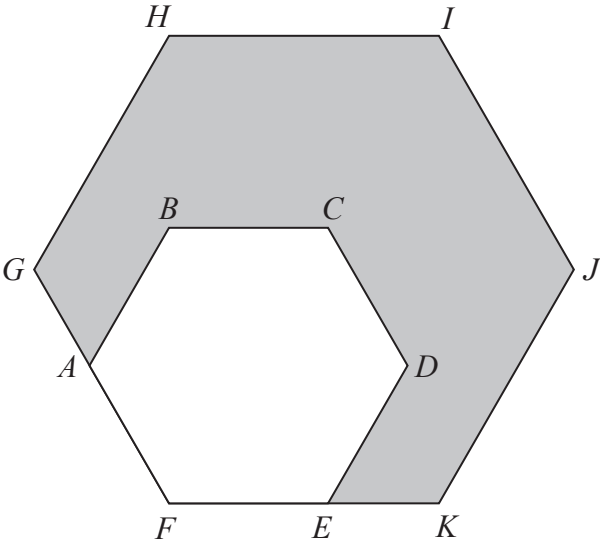




$ABCDEF$ is a regular hexagon with sides of length x .
This hexagon is enlarged, centre F , by scale factor p to give hexagon $FGHIJK$.
Show that the area of the shaded region in the diagram is given by $\frac{3\sqrt{3}}{2}(p^2 - 1)x^2$